Program:	MCA							Semester : II
Course:	Web Development using Django (PEC-1)							Code : MCA32PE02
Credits	Teaching Scheme (Hrs. / Week)			Evaluation Scheme and Marks				
	Lecture	Practical	Tutorial / Activity	FA		TW	SA	Total
				FA-1	FA-2			
2	2	-	-	10	10	-	30	50

Prior knowledge of :

1. Basic Programming Concepts
2. Basics of Web Development
3. Logical Thinking
4. Database Operations

is essential.

Course Objectives

This course aims at enabling students:

1. Understand the concept of dynamic web pages and the Django template system.
2. Learn how to map URLs to views and handle 404 errors in Django.
3. Gain knowledge of the Model-View-Template (MVT) development pattern in Django.
4. Develop skills in defining and installing database models using Django's ORM.
5. Explore the Django admin interface and its customization capabilities.
6. Learn about form processing, validation, and saving form responses in Django.
7. Understand access control mechanisms using sessions and user authentication in Django.
8. Implement login, logout, user management, and permission handling in Django applications.
9. Gain exposure to asynchronous messaging concepts

Course Outcomes

After learning the course, the students should be able to:

1. Create dynamic web pages using Django's template system and context objects.
2. Implement database models.
3. Implement Django admin interface for efficient management of application data, including users, groups, permissions, and form processing.
4. Implement the Django session framework to manage user sessions, handle session data in views, optimize session usage for performance, and ensure secure session management practices.

Detailed Syllabus

Unit No.	Description	Duration (Hrs.)
1	**Dynamic Web Pages And Django Template System :** Dynamic Content, Mapping URLs to Views, URL configurations and Loose Coupling, 404 Errors, Dynamic URLs, Django Template System Basics, Creating Template Objects, Rendering a Template, Rendering a Template, Playing with Context Objects	8
2	**Database Model With Django :**The MVT Development Pattern, Configuring Django for Database Access, Understanding Django Apps, Defining Models in Python, Installing the Model, Understanding Model	7

	Fields & Options, Creating A Django Model, Adding Model String Representations, Making Changes to a Database Schema.	
3	**Django Admin Interface :** Activating the Admin Interface, Using the Admin Interface, Users, Groups and Permissions Customizing the Admin Interface's Look and Feel, Customizing the Admin Index Page, When and Why to Use the Admin Interface, Form Processing, Form Validation Save Form Response	8
4	**Access Control With Session And Users :** The Django Session Framework, Sessions in Views, Session Tuning, Installing Django User Authentication, Using Authentication in Views, Login and Logout, Building your Own Login/Logout Views, Adding & Deactivating Users, Asynchronous Messaging, Managing Permissions	7
	Total	**30**

Text Books:

1. George, N. *Build a Website with Django 3*. Packt Publishing, (2020)

Reference Books:

1. Greenfeld, D. R., &Greenfeld, A. R. *Two Scoops of Django 1.11: Best Practices for the Django Web Framework*. Two Scoops Press, (2017)
2. Vincent, W. S. *Django for Beginners: Build Websites with Python and Django. WelcomeToCode*, (2021)

E-Resources:

1. **Django Tutorial :** https://www.w3schools.com/django/
2. **Learn Django Framework :** https://www.geeksforgeeks.org/django-tutorial/
3. **Django Tutorial :** https://www.tutorialspoint.com/django/index.htm